

M08 MODULE TEST - PROTECTED REVIEW

RC2 | Worked controls + diagnosis | Not a Gate answer key

Open only after a genuine saved Module Test attempt. Compare reasoning and units first; then compare numeric controls. Close the review before a changed retry.

Protected controls

Item	Expected control / interpretation
A1	$n=10$; mean=29.0; median=23.5; range=64. The 82-minute value pulls the mean upward much more than the median; investigate before deleting.
A2	$z=(66-50)/8=+2.0$. It is 2 SD above the mean under that standardization; z alone does not prove error or normality.
B1	Target=all monthly active users; frame=7-day app openers; selection/coverage bias. $n=5,000$ improves precision for that frame but does not repair systematic exclusion.
B2	$SE=18/\sqrt{144}=1.5$. Approx 95% CI= $72 \pm 2.94 = [69.06, 74.94]$. CI precision does not prove unbiased sampling.
B3	$SE=18/\sqrt{576}=0.75$; doubling $\sqrt{n}$ halves SE.
C1	Control=10.0%; Treatment=12.5%; absolute lift=+2.5pp; relative lift=+25.0%; $z=1.9380$; two-sided $p=0.0526$; 95% CI approx $[-0.026pp, +5.026pp]$.
C2	Support 6.0% vs 6.5%, change +0.5pp: no guardrail breach. Point estimate clears 2.0pp MWE, but CI lower bound does not; at alpha .05 do not claim a statistically established win. Continue/collect evidence or make a risk-based decision without overstating certainty.
C3	p is computed under H_0 ; it is not $P(H_0 \text{ is true})$.
D	Association wording only; plausible alternatives include prior ability/motivation/manager selection/job type. Stronger design: randomized training assignment where feasible.
E	Predeclare unit, eligibility, variants, primary metric, guardrail, duration/stopping rule, MWE, quality/sample-ratio check and decision rule before outcomes.

Diagnostic repair map

Miss pattern	Return only to
Mean/median/outlier or z-score misuse	STAT1-STAT2
Large-n fixes bias / SD-SE confusion / CI probability wording	STAT3-STAT5
p-value meaning / reject vs fail-to-reject	STAT6
pp vs relative %, MWE or guardrail reasoning	STAT7
Correlation described as causation	STAT8
Post-hoc metric/stopping changes or weak experiment plan	STAT9
Arithmetic right but decision ignores uncertainty/threshold/guardrail	M-STAT integration

Fresh retry contract

Repair the failed competency, close this review, then solve a changed example without reproducing these answers. Keep the original attempt and changed retry as separate evidence. Only fresh transfer upgrades mastery.